WEATHERING

1. Introduction to Weathering & Forces 🌍🌤️

Definitions:

- **Exogenic Forces:** Forces that operate on the *surface* of the earth ('exo' in Greek means outside) through processes such as weathering, mass wasting, erosion, transportation, deposition, and formation of soils.
- **Weathering:** The process of breaking down of rocks but *not* its removal. It is described as the disintegration or decomposition of a rock in size by natural agents at or near the surface of the earth.
    - *Handwritten Mastery:* Weathering strictly involves **"no movement"**; the rock's **"position remains the same"**! ✍️📍.

Key Points (Exhaustive Detail):

- The internal forces operating in the interior of the earth are described as 'Constructive Forces' because they lead to the formation of landforms.
- External (exogenic) forces derive their energy from the sun, with climate acting as the driving agent in all cases.
- Weathering is so called because the disintegration or decomposition of rocks takes place *only by being exposed*.
- It is the change in weather conditions like temperature, moisture, and precipitation that disintegrate or break down the rocks.
- Weathering is the basic process and an essential phase of denudation.

Things to Remember (The 7 Chief Characteristics of Weathering) 🧠

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    1. Weathering involves disintegration or decay of solid rocks.

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2. It depends on climatic elements and on the characteristics of rocks. For example, the rock's chemical composition, hardness, texture, and its permeability (allowing water to penetrate) determines the weathering pattern.

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3. Weathering affects the surface of the earth.

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4. Weathering causes formation of soil.

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5. Weathering involves not one but several processes to cause disintegration of rocks.

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6. Weathering changes hard massive rocks into finer materials.

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7. It prepares rock materials for transportation by agents of gradation.

The "Why?" Section ❓

- **Why can't weathering take place on the moon?** Weathering cannot take place on the moon's surface because the necessary weather conditions (temperature changes, moisture, precipitation) do not exist there! This is exactly why the footprints left on the moon by astronauts will last for millions of years.

2. Types of Weathering: Physical Weathering 🔨🪨

(Note: Weathering is a static process and is divided into three types: physical, chemical, and biological.)

Definitions:

- **Physical Weathering (Mechanical Weathering):** Involves rock disintegration *without any change in the chemical constituents* of the rock.
- **Block Disintegration:** The sudden rise and fall of temperature causes the sudden expansion and contraction of rocks, leading to tension and the splitting of rocks into blocks.
- **Granular Disintegration:** Weathering that reduces rocks made of different minerals into smaller pieces and fragments, often taking place in cold places and at high altitudes.
- **Frost Action:** When water collects in cracks during the day and freezes at night, it expands. This leads to the rupture of rocks on account of repeated actions of melting and freezing. *(Page-Shift Protocol Engaged! 🧩)*
- **Exfoliation:** Sudden contraction and expansion peels off the outer layer of rocks because they are not good conductors of heat.

Key Points (Exhaustive Detail & Handwritten Notes!):

- Main factors responsible for physical weathering are temperature changes, moisture, frost action, and wind.
- In block disintegration, the repeated action causes breakdown because the temperature falls suddenly at night and rises suddenly during the day.
- Sudden temperature changes may also cause fissures into which water penetrates, which in turn stimulates chemical weathering.
- **Handwritten Diagram Mastery:** You noted that the mechanics of physical weathering involve water collecting in cracks, and then freezing/expanding, which is specifically called **"Frost action"**! ✍️❄️.
- **Handwritten Mastery (Exfoliation):** You brilliantly simplified exfoliation: The **"outer layer"** experiences **"tension"**, which leads directly to **"onion peeling"**! ✍️🧅.

The "Why?" Section ❓

- **Why is physical weathering due to temperature changes more rapid in hot deserts?** It is because the temperature changes (the massive difference between hot days and freezing nights) are the sharpest in a desert!.

3. Chemical Weathering 🧪🪨

Definitions:

- **Chemical Weathering:** When some or all of the minerals in rocks suffer decay or alteration by agents like water, oxygen, carbon, or organic acids, transforming them into different chemical combinations.
- **Oxidation:** The reaction of minerals in rocks to atmospheric oxygen (which is also contained in rainwater).
- **Carbonation:** When mineral constituents of rocks (like carbon dioxide) come in contact with water and produce an acidic effect on rocks.
- **Hydration:** The expansion of minerals that occurs on coming into contact with rainwater, causing them to become heavy and start disintegrating.
- **Solution:** The process in which minerals in the rocks dissolve completely in water.

Key Points (Exhaustive Detail & Handwritten Notes!):

- Chemical weathering leads to the true *decomposition* of rocks and minerals.
- **Oxidation Effects:** On coming in contact with iron-compounds in rocks, the iron starts rusting or crumbling, and the colour of the rock changes to red, yellow, or brown.
- **Carbonation Effects:** This action dissolves most calcareous rocks such as gypsum, marble, and limestone. Rainwater converts calcium carbonate into calcium bicarbonate, which is soluble and is taken away by ground water. (Note: This must not be mistaken for the "solution" process!).
- **Hydration Effects:** Though it can be termed as physical or mechanical, its effect is a chemical process. Rocks like feldspar are converted into kaolin.

- **Solution Effects:** All rocks are subjected to solution to some extent. The rate is affected by mineral composition AND structure. (e.g., Sedimentary rocks have pore-spaces where air and water can lodge and attack).
    - *Handwritten Mastery:* You noted that this process directly leads to **"dissolved minerals"**! ✍️💧.
    - *Example:* Rainwater causes chemical disintegration of rock-salt and gypsum.

4. Biological Weathering 🌱🐜

Definitions:

- **Biological Weathering (Organic Weathering):** Weathering where the main agents are animals, insects, plants, and man.

Key Points:

- In all biological cases, both physical disintegration and/or chemical decomposition are involved.

The "Why?" Section ❓

- **Why does biological weathering always involve chemical decomposition?** This is because all biological matter is made up of organic compounds, and these compounds chemically react with the minerals present in rocks!.

5. Table of Differences 🆚

(Extracted flawlessly from Table 9.1)

Feature	**Mechanical (Physical) Weathering** 🔨	**Chemical Weathering** 🧪
1. Primary Action	Disintegration of rocks takes place *without* any change in chemical constituents of rocks.	Decomposition of rocks takes place *with* changes in the chemical constituents of rocks.

Feature	Mechanical (Physical) Weathering 🔨	Chemical Weathering 🧪
2. Causal Factors	Factors such as temperature, moisture, frost action, and wind cause the physical break-up of rocks.	Temperature, moisture, etc., cause minerals in rocks to either dissolve in water or change their composition.
3. Main Agents	Its agents are temperature and moisture.	Its agents are water, oxygen, carbon, and various organic acids.
4. End Result	It increases the surface area, making rocks vulnerable to chemical weathering.	As its result, new minerals are formed or existing ones dissolve, leading to softer, weaker rock.

4. Biological Weathering (The Main Agents) 🌱🐜

(Continued seamlessly from the previous page!) 🧩

Key Points (Exhaustive Detail & Handwritten Notes!):

- **(i) Humans:** Man is the most active agent of *both* physical and chemical weathering of rocks.
    - *Physical:* Mining, excavations, and the construction of roads and buildings directly cause the physical disintegration of rocks. ⛏️
    - *Chemical:* Agriculture and the dumping of chemicals and wastes underground actively promote both physical and chemical weathering. 🛢️
- **(ii) Plants:** Roots of large trees reach deep into rocks and cause physical disintegration due to immense pressure.
    - *Handwritten Mastery:* You drew a clear arrow here noting how these **"roots penetrate"** directly into the rock layers to break them apart! ✍️🌳
    - *The Exception:* Most vegetation, however, actually *prevents* disintegration of rocks. It binds the surface layer together and

strictly does not allow the exposure of rocks beneath to the elements of weathering. 🛡️

- **(iii) Animals and Insects:** Animals like rabbits, worms, moles, and insects bring large quantities of fine material up to the surface. 🐇
    - Burrowing animals help to loosen the surface materials around the rocks, severely facilitating their physical disintegration. 🕳️
    - Upon death, the decaying bodies of animals also provide many chemicals and acids for chemical rock disintegration. ☠️🧪

The "Why?" Section ❓

- **Why does vegetation sometimes *prevent* weathering instead of causing it?** Because dense plant roots bind the top surface soil and rock fragments tightly together, acting as a protective shield that physically blocks the underlying rocks from being exposed to harsh weather elements! 🛡️🌿

5. Weathering in Different Climatic Regions ⛈️🌍

Key Points (Exhaustive Detail):

- From all the processes discussed, it is absolutely clear that *climate is the most important factor* in weathering.
- As a general rule: Physical weathering is more rapid in desert climates, and chemical weathering occurs in moist and cold climates. 🏜️❄️

Breakdown by Region:

- **(i) In Equatorial Climates:** Humidity and temperature are constantly high. Weathering, especially *chemical weathering*, is highly active in these regions. 🌴🧪
- **(ii) In Tropical Climates:** There is a marked dry season and wet season. Differences in heating lead to a consistently high rate of evaporation.
    - *Soil Formation:* During the wet season, the precipitation of oxides of iron and aluminium takes place from rocks. This directly promotes the formation of *laterite soils*. ⛈️🪨

- **(iii) In Dry Climates or Deserts:** Mechanical (physical) weathering is the most dominant strictly due to the high *diurnal (daily) range of temperature*. 🐫🌡️
- **(iv) In Mid-latitudes or Temperate Climates:** The absolute most powerful agent of weathering here is *frost action*. However, in limestone areas, 'Solution' (chemical weathering) operates on a massive scale. ❄️
- **(v) In Polar Climates:** Physical weathering due to frost action is far more prevalent, while biological weathering is completely negligible. 🧊

The "Why?" Section ❓

- **Why is chemical weathering surprisingly active in freezing Polar regions?** Recent studies have proven that the melting water of polar regions contains *more carbonic acid content*! This happens because carbon dioxide is scientifically *more soluble at low temperatures* than at high temperatures, turning the freezing water into a highly active chemical agent! ❄️🧪